Embedded Control for Robotics

Lecture # 02 Control Fundaments

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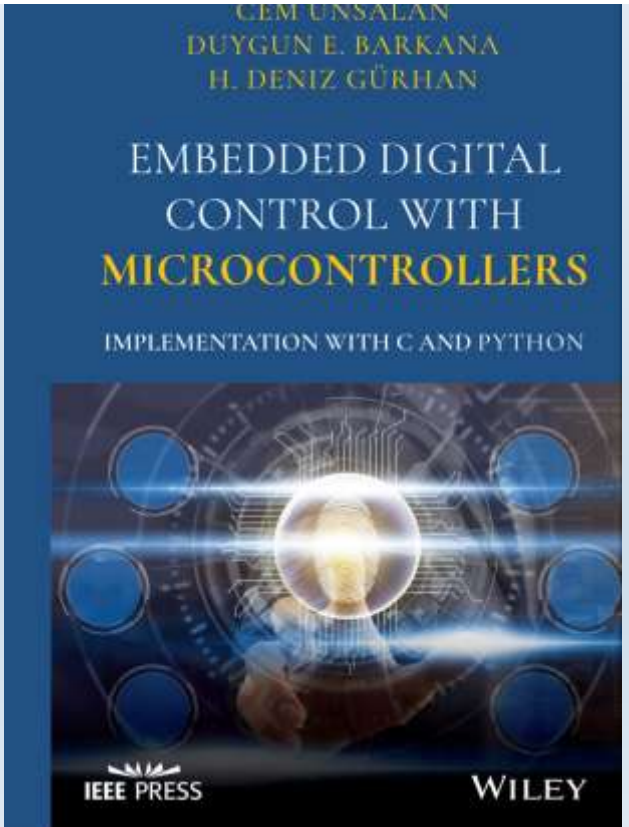
Course Code

Course Code	CS-4095
Course Title	Embedded Control of Robotics
Credit Hours	3+0

Books

Textbook	1. Embedded Digital Control with Microcontrollers, 1st Edition, 2021, Cem Unsalan Duygun E. Barkana, H. Deniz Gurhan, Wiley-IEEE Press
Reference Material	1. Classical and Modern Controls with Microcontrollers, Design, Implementation, and Applications, Springer, 2019, Ying Bai, Zvi S. Roth 2. Process Control in Practice, 1st Edition, 2023, Tore Hägglund, Walter de Gruyter GmbH





Course Assessment

Assessment Instruments with Weights (homework, quizzes, midterms, final, programming assignments, lab work, etc.)

100% Theory

Assessment Items

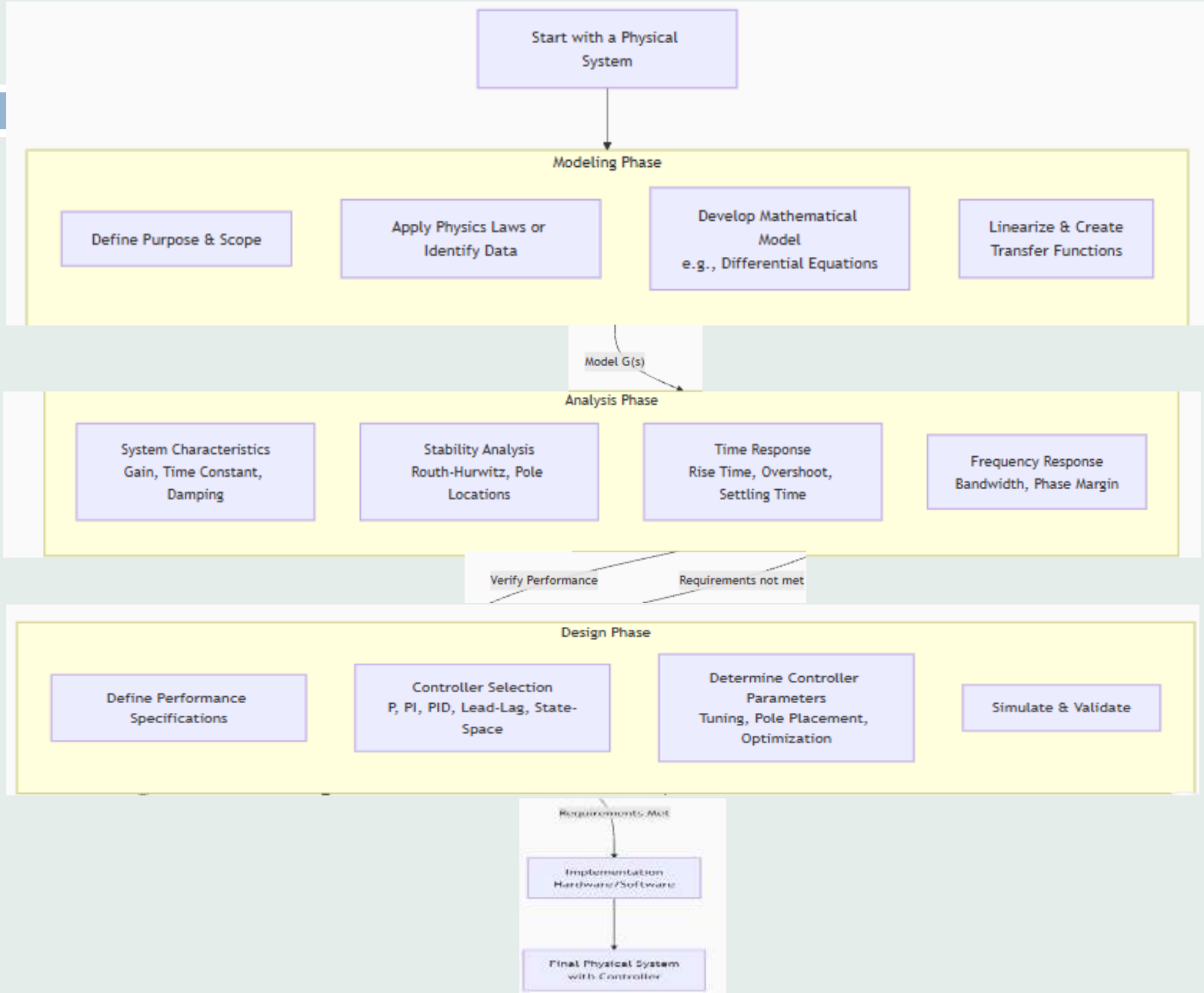
Assessment Item	Number	Weight (%)
Assignments	3-5	10
Project	1	10
Sessional 1	1	15
Sessional 2	1	15
Quizzes	4-6	10
Final Exam	1	40

Control Learning philosophy

- It is about *your learning* - for which *You* is the boss!
- We employ feed forward and feedback to improve the learning process
- *Feed forward* concepts: material, lectures, the project etc.
- *Feedback* mechanisms
  1) Participant representatives: to collect important feedback info
  2) Lecture and workshop follow up questions
  3) Hand ins of project parts
  4) Course evaluation in the end

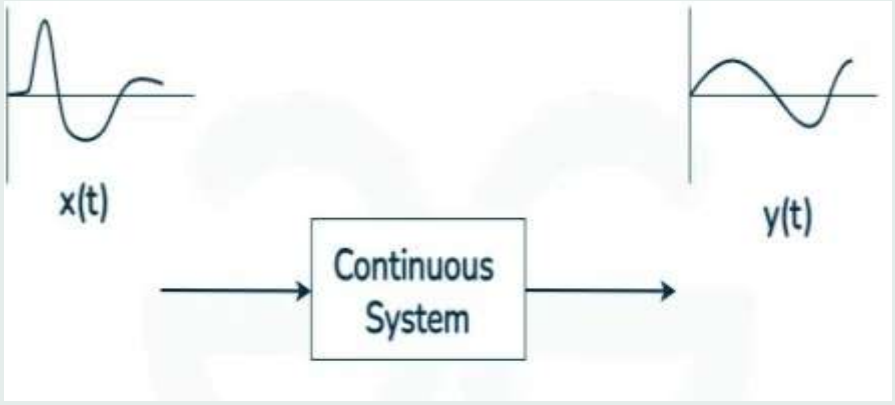
Control Systems: Big Picture

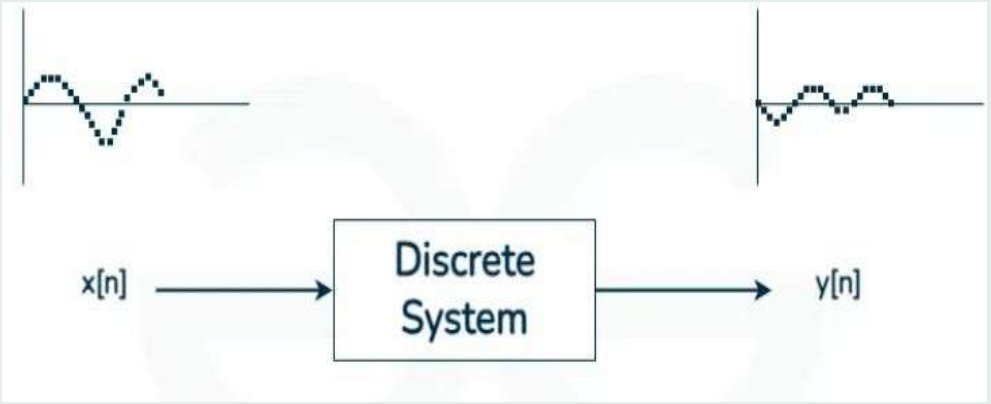
1. Modeling
2. Analysis
3. Design



Control Systems: Simplified Picture

Continuous Time		Discrete Time	
Differential Equations		Difference Equations	
Laplace Transform, Transfer Function	State Space Representation	Z transform, Transfer Function	State Space Representation
Root Locus Analysis	State Space: Modeling, Analysis, Design	Root Locus Analysis	State Space: Modeling, Analysis, Design





Continuous time

Differential equations (time)

$$\dot{y} = -ay + bu$$

State space models (time)

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$

Transfer functions (frequency)

$$y(s) = \frac{b}{s+a}u(s)$$

Discrete time

Difference equations (time)

$$y[n] = ay[n-1] + bu[n]$$

State space models (time)

$$x[n+1] = \Phi x[n] + \Gamma u[n]$$
$$y[n] = Cx[n] + Du[n]$$

Transfer functions (frequency)

$$y(z) = \frac{b}{z+a}u(z)$$

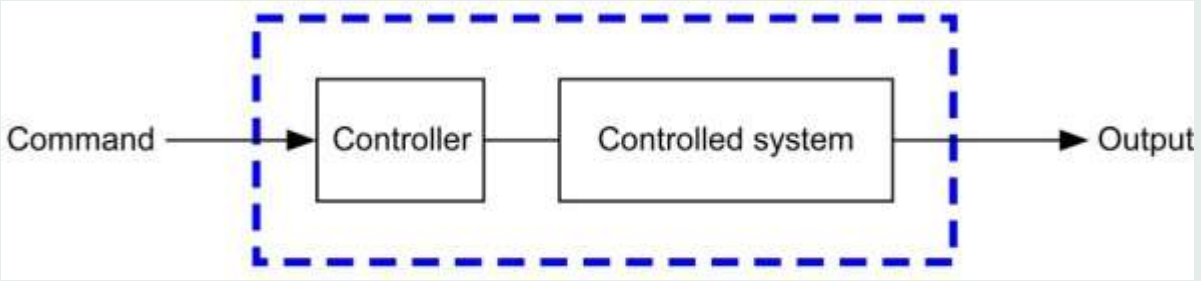
System/Process

- A system can be defined as an integrated body or an element. For example,
  - a vehicle
  - a motor
  - a human being body
  - or company
  - even our Earth, can be considered as a system.
- A system can also be called a process or a plant.



Control System

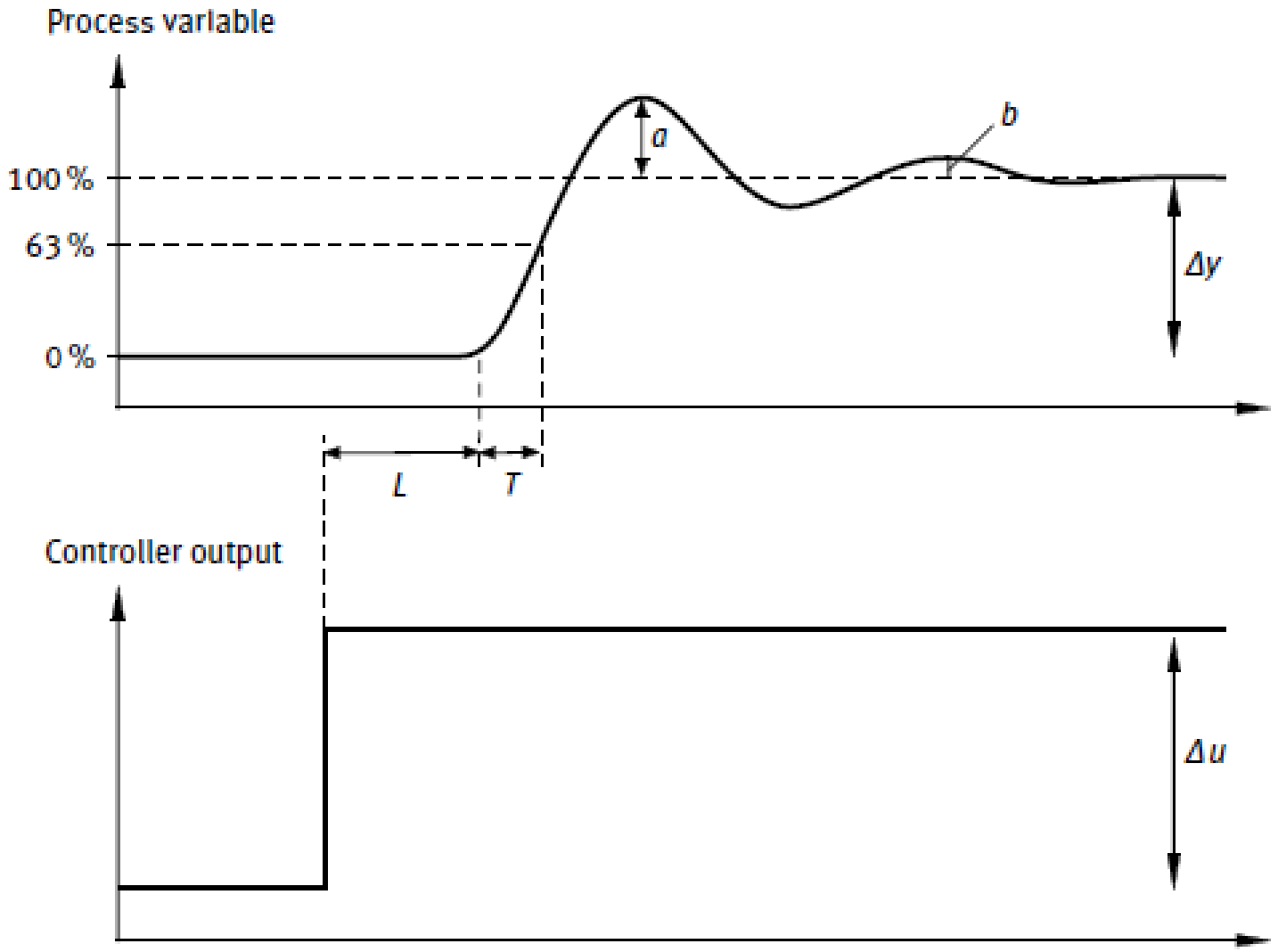
- A control system should contain a controller to perform various control functions to the system to get the desired outputs. It Consist of
  - inputs,
  - controller,
  - system or a process,
  - outputs.



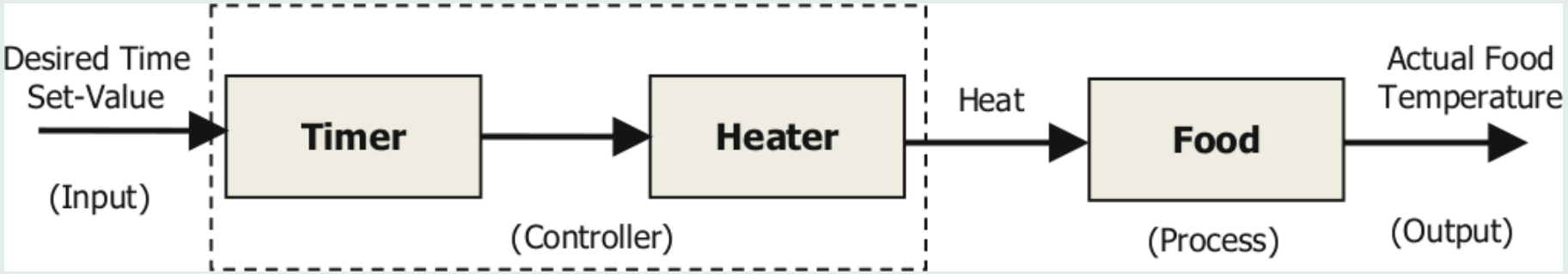
Open Loop Control System

- The output is neither measured nor feedback for comparison with the input
- An open-loop system is expected to faithfully follow its input command or set point regardless of the final result
- This control system is also called a non-feedback control system

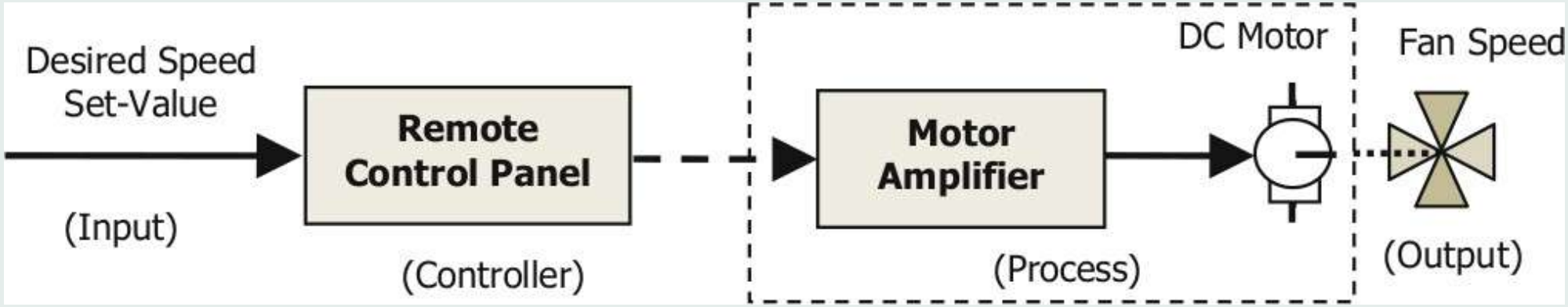
Output response



Open-Loop Control System Microwave



Open Loop Control System DC Motor



Open-Loop Control System Problems

1. **Inaccuracy and Sensitivity to Disturbances**
   There is no comparison between actual input and desired output values.
2. **In-ability to Adapt to Changing Conditions**

   Has no self-regulation or self-control action over the output value.

   Cannot reduce or overcome all variations or disturbances coming from the external condition
3. **No Error Detection**

   The system is "blind" to its own performance. It has no way of knowing if it has succeeded or failed in its task.

Example: An automatic coffee maker pours a pre-measured amount of water. If the water reservoir is empty, it will still "go through the motions" and produce no coffee, with no indication of the failure.

When Are They Used?

- The system is simple and well-understood.
- Disturbances are either negligible or predictable.
- The cost of implementing feedback (sensors, complex controllers) is not justified.
- The task is repetitive and high speed is required (e.g., in manufacturing).

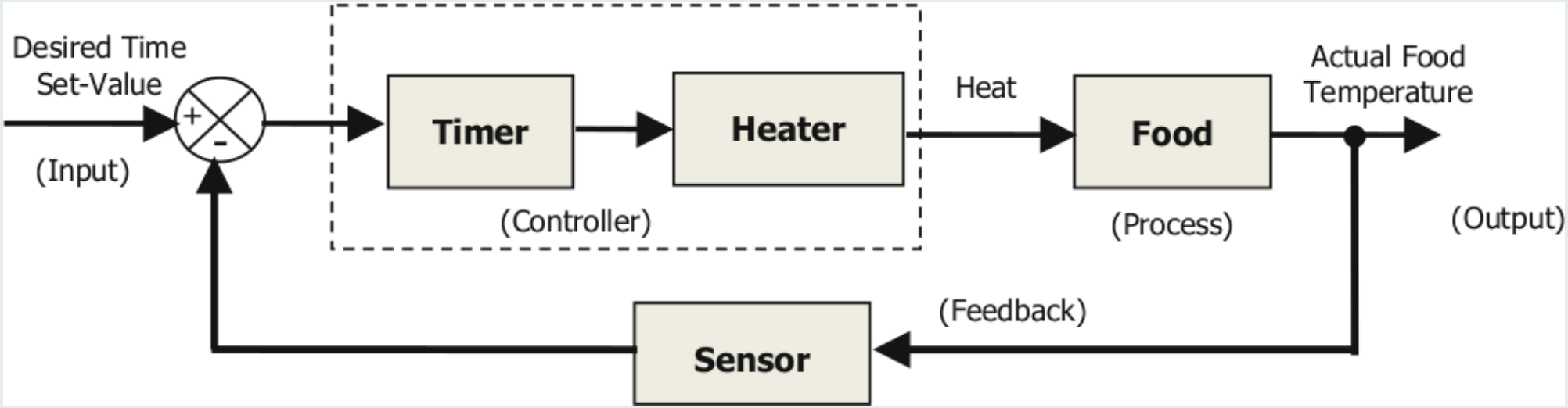
Closed-Loop Control System

- Feedback Control System
- It uses the concept of an open-loop system as its forward path PLUS has one or more feedback loops between its output and its input.
- Use of a feedback control in order to reduce any errors between the output and the input

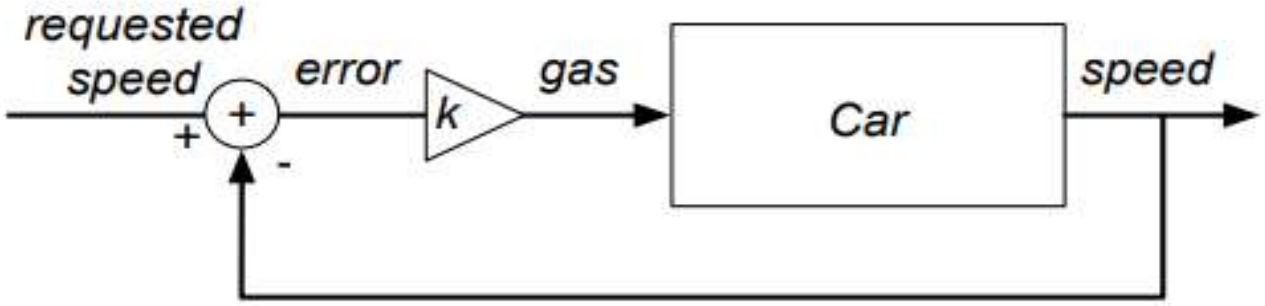
Test

- Closed-loop control system for
  - Microwave
  - DC Motor

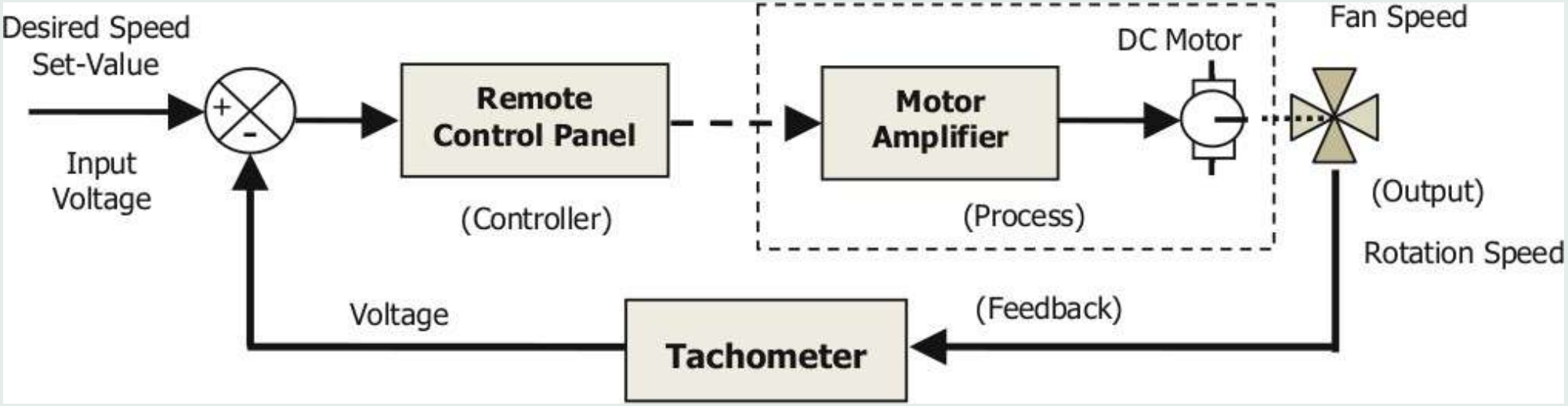
Closed-Loop Control System Microwave



Use feedback by comparing the measured speed to the requested speed:



Closed-Loop Control System DC Motor

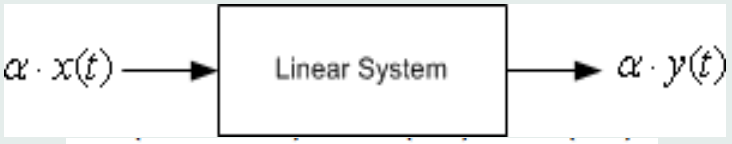


Closed-Loop Control System Advantages

- Reducing errors by automatically adjusting the systems input.
- Improving stability of an unstable system.
- Reducing the systems' sensitivity to enhance robustness against external disturbances or internal variations to the process.
- Producing a reliable and repeatable performance.

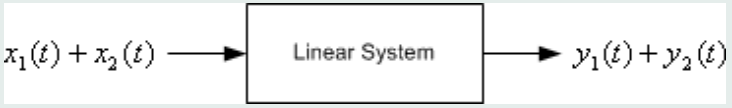
Linear System & Laws

- The relationship between the input and output is linear
- If you were to double the input, the output would also double, assuming everything else remains constant
- Homogeneity



$$f(c \cdot x_1) = c \cdot f(x_1)$$

- Additivity



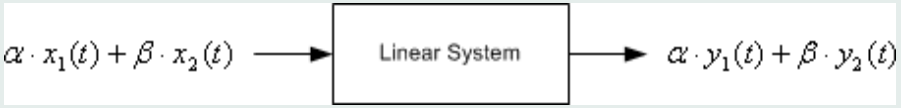
$$f(x_1 + x_2) = f(x_1) + f(x_2)$$

Exp: simple circuit with just resistors and a battery that gives a steady flow of electricity

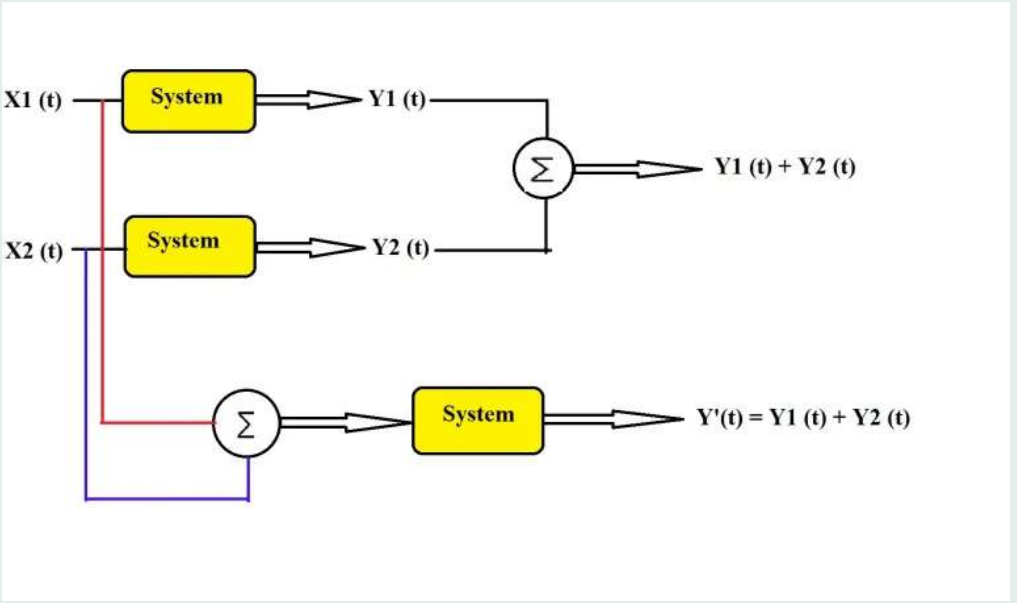
Principal of Superposition

- Principle of superposition applies: the response caused by multiple inputs can be found by summing the individual responses to each input

Superposition



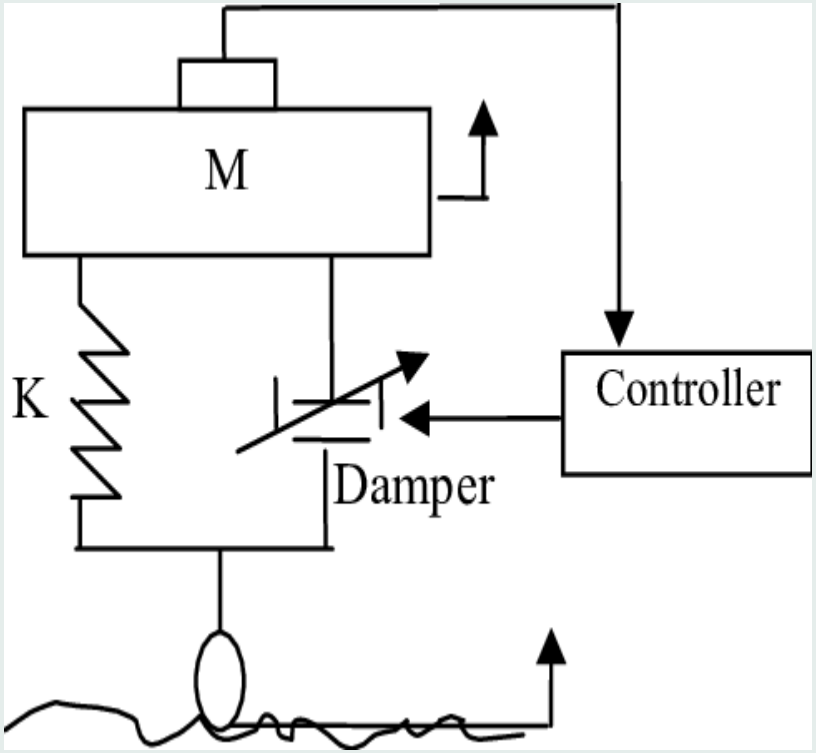
$$f(x_1(t)) = y_1(t)$$
$$f(x_2(t)) = y_2(t)$$
$$f(x_1(t) + x_2(t)) = y_1(t) + y_2(t)$$



Non-Linear System

- Complex, non-proportional response to inputs
- Require specialized methods for analysis and control

  Examples: diodes, transistors, predator-pray, market, biological systems



Causal System

- A system whose output or response at any time instant (t) depends only on the present and past values of the input but not on the future values of the input is called the causal system.
- the output or response does not begin before the input signal is applied. This is why, a causal system is also called a non-anticipated system.

$h(t) = 0; t < 0$ ... for continuous time system

$h(n) = 0; n < 0$ ... for discrete time system

Non-causal System

- A system whose output or response at any time instant (t) depends upon future values of the input is called the non-causal system.
- These systems can generate an output before the input is given, hence they are also known as **anticipative systems**.

$$y(t) = x(t+1)$$

$$y(n) = x(n+1)$$

Causal/Non-causal systems

$$y(t) = x(t - 3) + 3x(t)$$

$$y(t) = tx(t)$$

$$y(n) = x(n - 1) + x(n)$$

$$y(n) = nx(n)$$

$$y(t) = x(t + 3) + 2x(t)$$

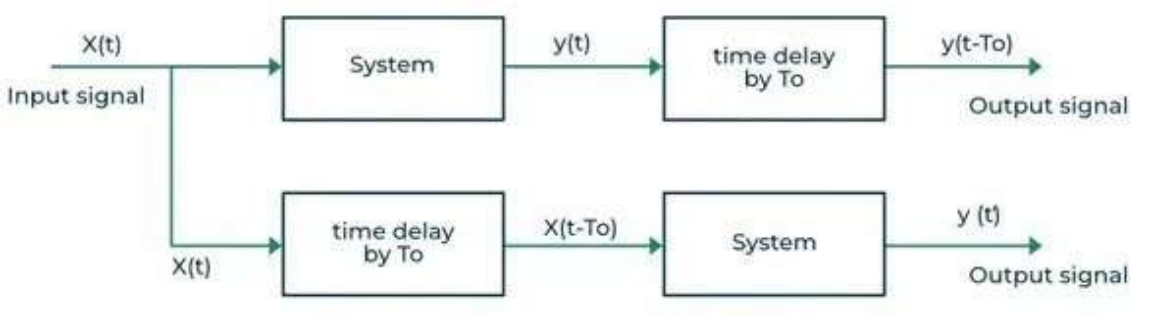
$$y(t) = x^2(t) + tx(t + 3)$$

$$y(n) = x(n) + 3x(3n + 3)$$

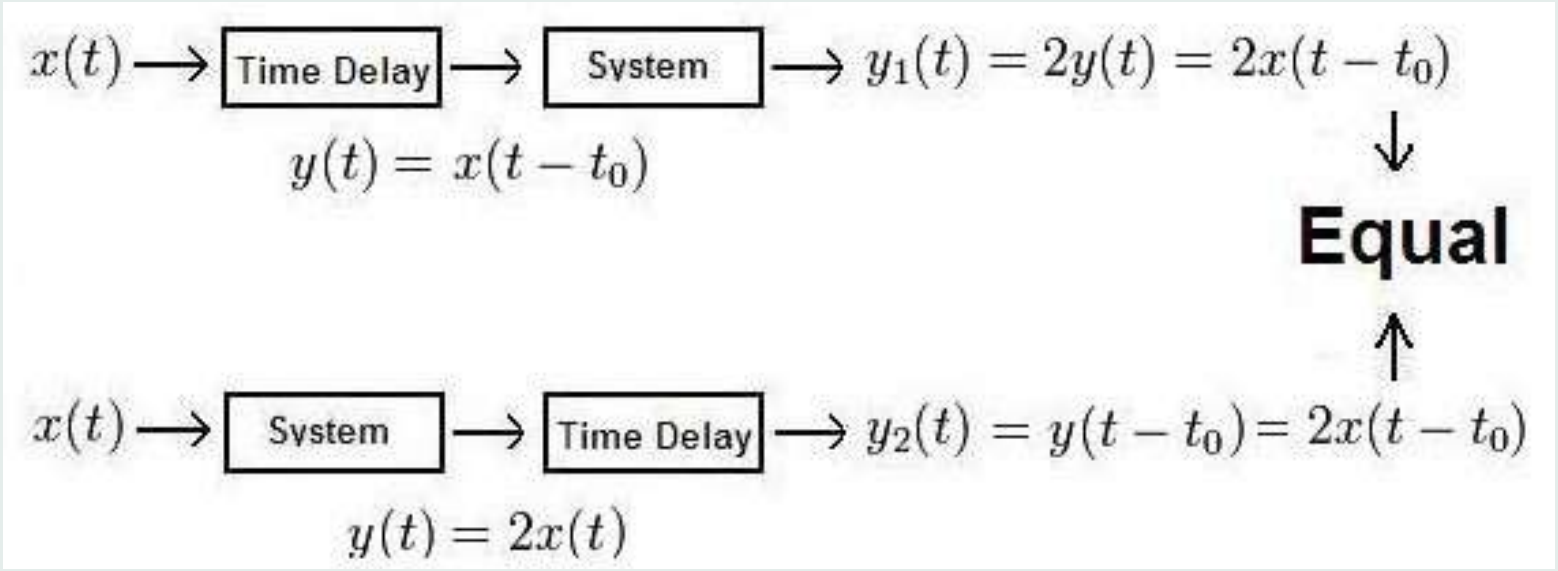
$$y(n) = x^2(n) + 2x(n + 1)$$

Time-Invariance

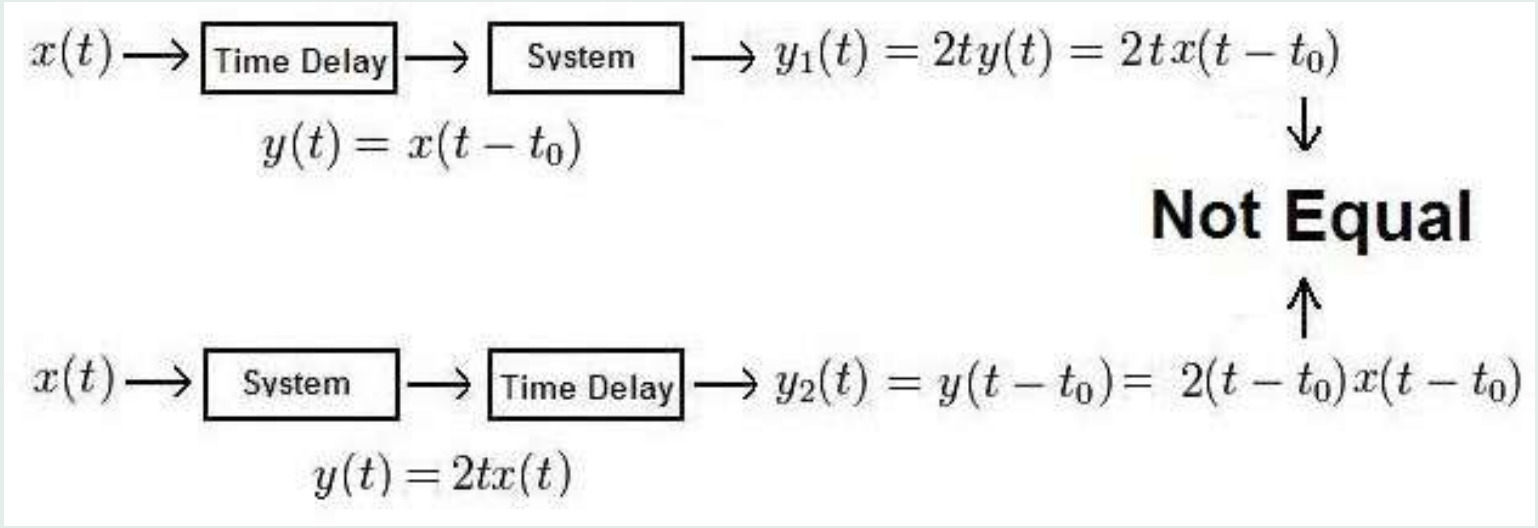
- A system is time-invariant if a time shift in input signal causes an identical time shift in output signal.



Time Invariant



Time Variant



Laplace Transforms

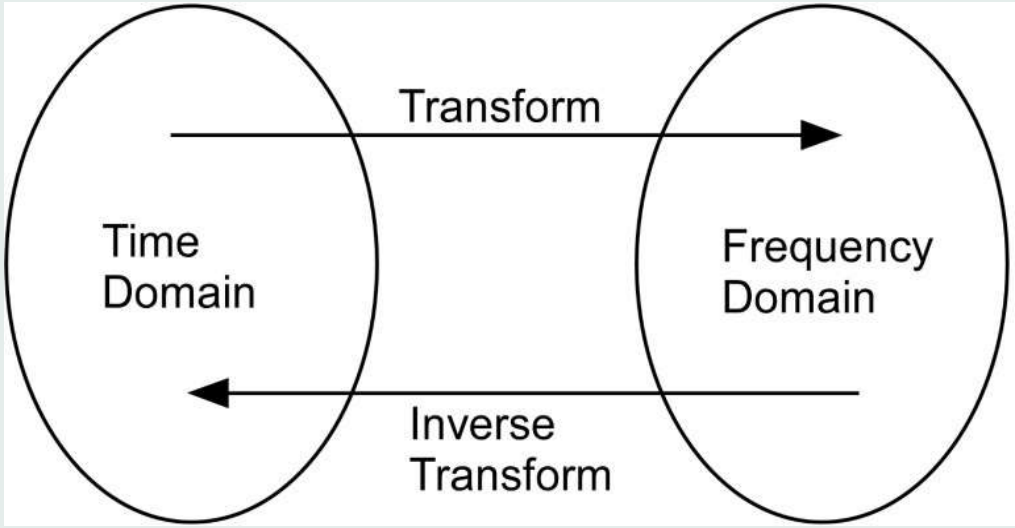
The Laplace transform of a time serie *u(t)* is defined as:

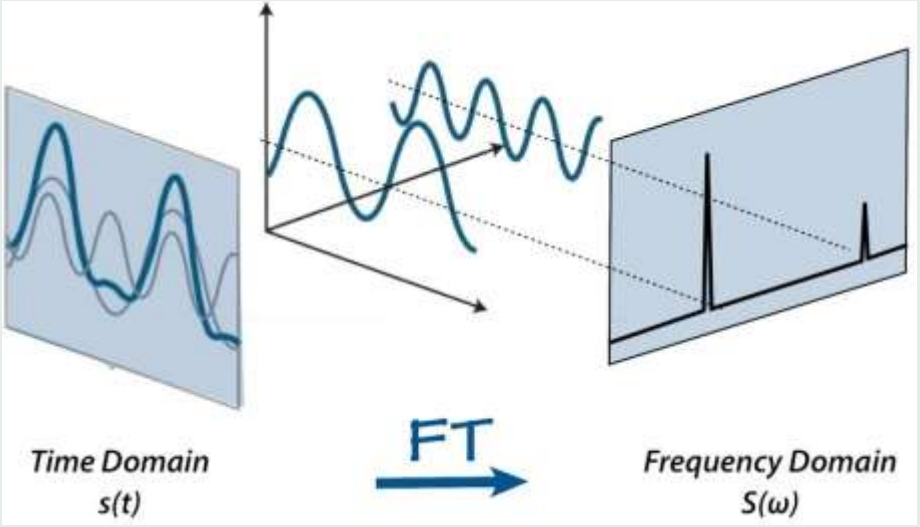
$$L\{u(t)\} = \int_0^\infty u(t)e^{-st}dt$$

A transfer function *G(s)*, is the ratio of the output Laplace transform with the input Laplace transform.

$$G(s) = \frac{L\{y(t)\}}{L\{u(t)\}}$$

$$Y(s) = G(s)U(s)$$

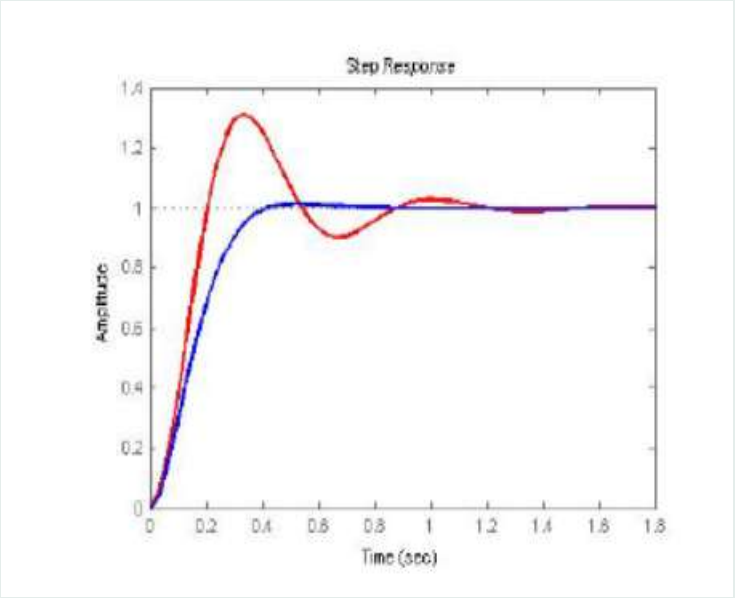




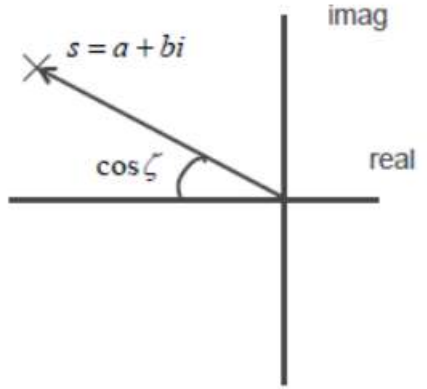
- Time domain
  - The response $y = G(s)u$ for different u, e.g., step, ramp, etc.

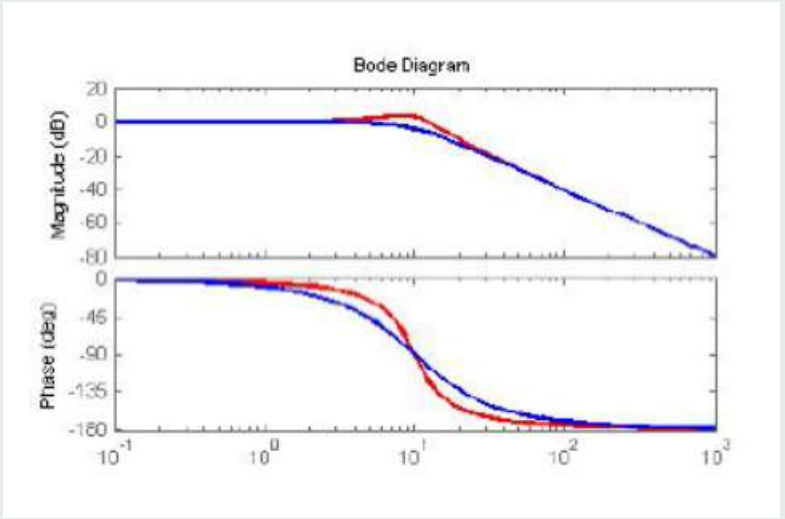
- Complex pole-zero plane
  - Solve for s in numerator and denominator polynomials

- Frequency domain
  - $G(s) = G(j\omega)$, ω is the frequency



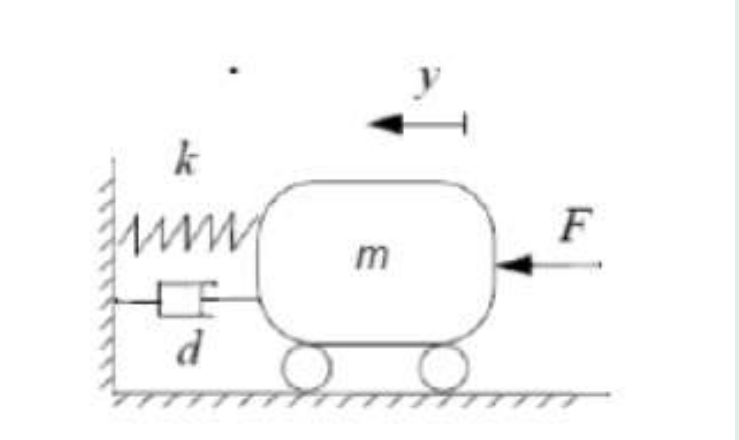
Poles and zeros can be plotted in the complex plane, the real part vs the imaginary part





Exp: Transfer function

Direct calculation using 2nd law of Newton to derive differential equation



$$L\{m\ddot{y} = F - d\dot{y} - ky\}$$

$$ms^2 y = F - dsy - ky$$

$$(ms^2 + ds + k)y = F$$

Taking Laplace transform

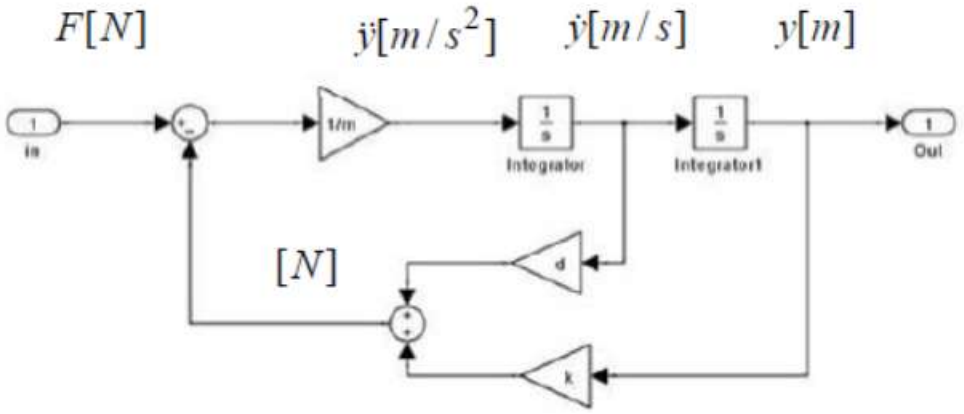
$$G(s) = \frac{F}{y} \qquad G(s) = \frac{1}{(ms^2 + ds + k)}$$

Block diagram in Simulink-Matlab

Differential equations are modeled by using integrators

$\frac{1}{s}$

$$\ddot{y} = \frac{1}{m}(F - ky - d\dot{y})$$

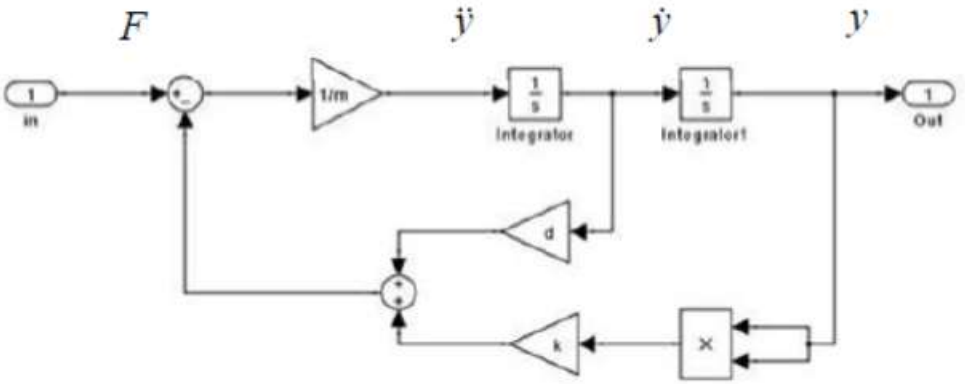


The signals have real units, force, position etc. for increased understanding. It is a specification that you also can simulate.

Non Linear model

Model of a nonlinear spring

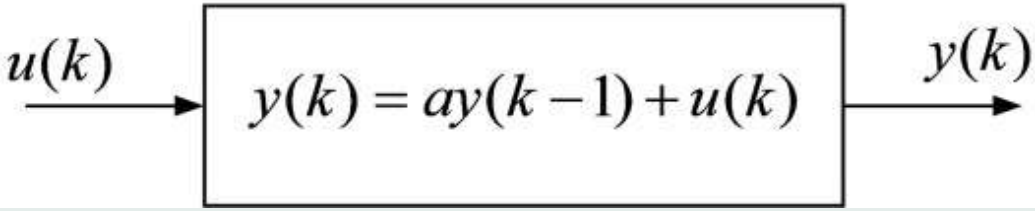
$$\ddot{y} = \frac{1}{m}(F - ky^2 - d\dot{y})$$



Difference Equations

- Difference equations are the discrete equivalent of differential equations
- Recursively defined sequences used to model discrete-time systems

$$y(k) = ay(k-1) + u(k)$$



Difference Equation

-
  - Each year, 1000 salmon are stocked in a creak and the salmon have a 30% chance of surviving and returning to the creak the next year. How many salmon will be in the creak each year and what will be population in the very far future?

$$y_{n+1} = 0.3y_n + 1000.$$

Assignment # 1

- DC motor Model
- Transfer function as Voltage input and Position OutPut
- Transfer Function as Volatge input Velocity output
- Next week Wednesday 3rd September 2025

References

- Classical and Modern Controls with Microcontrollers, Ying Bai and Zvi S. Roth